

WBAG: A Whole-Body and Attached-Geometry Safety Framework for Vision-Language-Action Manipulation

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Abstract—Vision-language-action (VLA) policies have demonstrated impressive capabilities in generalizable robotic manipulation, but their deployment in the real world remains challenging due to potential collisions involving different parts of the robot, manipulated objects, and the surrounding environment. Existing inference-time VLA safety frameworks typically rely on simplified end-effector-centered representations that do not explicitly model the full articulated robot and attached-object geometry. In this paper, we present WBAG, a safety framework that models the robot’s whole-body and grasp-dependent attached geometry. WBAG constructs a grasp-conditioned safe set that adapts the protected geometry as objects are grasped, then converts this evolving geometry into differentiable CBF constraints that minimally modify the VLA’s native six-dimensional operational-space action for collision avoidance across robot, scene, and attached geometry. On the SafeLIBERO benchmark, a variant of LIBERO augmented with obstacles for safety evaluation, WBAG achieves the best overall safety and safe task success among the evaluated methods under a scene-level safety evaluator that monitors all eligible non-task objects, reaching 97.38% aggregate Scene Safety and 59.38% Safe Success.

I. INTRODUCTION

Vision-language-action (VLA) models map visual observations and language instructions directly to robot actions across diverse manipulation tasks [1], but their task-level generalization does not provide an explicit geometric safety model to consistently safeguard robot motions. In cluttered environments, a VLA action may move the end effector toward the target while causing an upstream link, wrist, gripper, or manipulated object to collide. The relevant collision geometry spans the articulated robot and changes when the robot grasps an object. Recent benchmark evaluations further show that task completion alone can obscure physical safety violations during manipulation [2], [3].

Control-theoretic safety methods, particularly control barrier functions (CBFs), provide a principled mechanism for enforcing collision avoidance during robotic control [4]. CBFs constrain control inputs to maintain forward invariance of a prescribed safe set, but their effectiveness depends on how that set is represented. For geometric collision avoidance, the safety function must capture separation between the protected robot geometry and surrounding obstacles.

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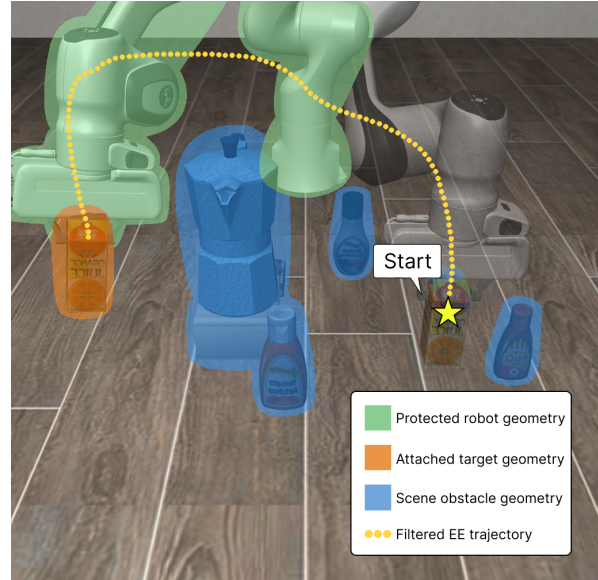


Fig. 1. Example of our WBAG execution. Green shows the constructed protected whole-body robot geometry, and orange shows the attached target geometry after grasping, with blue showing the extracted scene obstacle geometry. The yellow dotted curve is the filtered safe end-effector trajectory.

The geometric representation therefore determines which collision modes can be enforced.

Recent work on safety-critical control with VLA-based policies [5]–[9] spans both training-time and inference-time interventions. SafeVLA [5] incorporates safety constraints during policy optimization through constrained reinforcement learning. At inference time, AEGIS [6] combines perception-derived geometric information with CBF constraints to filter a pretrained VLA, while KNOWS [7] combines VLA-based target identification with online obstacle tracking. Other methods enforce safety directly within flow-matching action generation. Recent work in [8] apply predictive trajectory-level barrier constraints through minimum-norm corrections during iterative denoising, while [9] modify the flow-matching dynamics using an aggregate CBF over the generated action chunk. However, existing inference-time safety frameworks typically enforce collision avoidance using simplified end-effector-centered geometry [6]–[8]. Such end-effector-centered representations do not explicitly model the full articulated robot or the rigid geometry of an attached object: the end effector may remain separated while an upstream link, wrist, or attached geometry collides with an obstacle.

Geometric methods such as Bernstein-polynomial signed distance fields (BP-SDFs) [10] provide smooth, differen-

tiable representations of object geometry, enabling efficient distance queries and gradients that are well suited for optimization-based control. For articulated robots, BP-SDFs can represent individual links, whose poses are determined by forward kinematics as a function of robot configuration. Recent work [11] employs BP-SDFs to construct geometry-aware CBFs from the minimum distance between robot and obstacle geometries and derive corresponding safety constraints on the robot control input. Applying such geometric safety constraints to VLA manipulation requires connecting perception, manipulation state, and the policy's native action interface. In our setting, scene and target geometry are reconstructed from RGB-D observations rather than assumed known, while the protected geometry changes when the target becomes rigidly attached after grasping. At the same time, the resulting link- and object-level clearances are functions of robot configuration, whereas the frozen VLA produces six-dimensional operational-space actions. These factors must therefore be coupled so that perception-derived, grasp-conditioned geometric constraints can be enforced directly on the VLA action without retraining the VLA or modifying the operational-space controller.

In this paper, we introduce **WBAG**, a whole-body and attached-geometry safety framework for pretrained VLA policies. WBAG reconstructs scene and target geometry from RGB-D observations and constructs a **grasp-conditioned safe set**: before grasping, it protects the articulated robot; after grasping, the protected set expands to include the rigidly attached object. We represent robot, scene, and attached geometry with differentiable BP-SDFs and enforce the resulting CBF constraints directly on the VLA's native six-dimensional operational-space action. WBAG minimally modifies the policy output while accounting for collisions involving the whole robot and attached geometry, without retraining the underlying VLA or modifying the operational-space controller. We evaluate WBAG on SafeLIBERO using a scene-level safety evaluator that monitors all eligible non-task scene objects. WBAG achieves the highest aggregate Scene Safety and Safe Success among the evaluated configurations, reaching 97.38% and 59.38%, respectively. Our main contributions are summarized as the following:

- We formulate a **grasp-conditioned geometric CBF** over articulated robot links and dynamically attached geometry, with the protected set adapting across pre- and post-grasp manipulation.
- We develop **WBAG**, a novel perception-to-control safety framework that reconstructs scene and target geometry from visual observations, maintains grasp-conditioned protected geometry, and enforces the resulting constraints directly on the VLA's native operational-space action without retraining the VLA or modifying the operational-space controller used to execute its actions.
- We conduct a **controlled SafeLIBERO evaluation** with scene-level safety measurement that isolates the effects of scene coverage, whole-body protection, and attached-

geometry modeling on safe VLA manipulation.

II. PRELIMINARIES

A. Control Barrier Functions

Consider the control-affine system

$$\dot{x} = f(x) + g(x)u, \quad (1)$$

where $x \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ is the system state, $u \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$ is the control input, and $f : \mathcal{X} \rightarrow \mathbb{R}^{n_x}$ and $g : \mathcal{X} \rightarrow \mathbb{R}^{n_x \times n_u}$ are locally Lipschitz continuous. Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be continuously differentiable and define the safe set

$$\mathcal{C} = \{x \in \mathcal{X} : h(x) \geq 0\}. \quad (2)$$

Following the standard CBF formulation [4], h is a control barrier function if there exists an extended class- $\mathcal{K}$ function α such that

$$\sup_{u \in \mathcal{U}} \{ \nabla_x h(x)^\top (f(x) + g(x)u) \} \geq -\alpha(h(x)), \quad (3)$$

for all $x \in \mathcal{X}$. Accordingly, the admissible control set is

$$\mathcal{B}(x) = \{u \in \mathcal{U} : \nabla_x h(x)^\top (f(x) + g(x)u) + \alpha(h(x)) \geq 0\}. \quad (4)$$

Any Lipschitz continuous controller satisfying $u(x) \in \mathcal{B}(x)$ renders $\mathcal{C}$ forward invariant.

For geometric collision avoidance, h can be defined from robot–obstacle clearance.

B. Bernstein Polynomial Signed Distance Fields

We represent rigid-body geometry using Bernstein polynomial signed distance fields (BP-SDFs) [10], [11]. For normalized local coordinates $\xi \in [0, 1]^3$, let $B(\xi) \in \mathbb{R}^{Q^3}$ denote the tensor-product Bernstein basis of order Q . The local-frame BP-SDF of a rigid body b is the scalar-valued function $\bar{\phi}_b : [0, 1]^3 \rightarrow \mathbb{R}$,

$$\bar{\phi}_b(\xi) = c_b^\top B(\xi), \quad (5)$$

where $c_b \in \mathbb{R}^{Q^3}$ contains the fitted Bernstein coefficients. The value $\bar{\phi}_b(\xi) \in \mathbb{R}$ approximates the signed distance to the surface of body b at the spatial location represented by ξ . We use the convention that $\bar{\phi}_b > 0$ outside the body, $\bar{\phi}_b = 0$ on its surface, and $\bar{\phi}_b < 0$ inside the body.

For an articulated robot, let $q \in \mathbb{R}^{n_q}$ denote the robot configuration. Forward kinematics determines the pose of each robot link b as a function of q . For a world-frame query point $x \in \mathbb{R}^3$, let $\xi_b(x; q) \in [0, 1]^3$ denote its normalized local coordinates in the frame of link b . The corresponding world-frame BP-SDF is

$$\phi_b(x; q) = \bar{\phi}_b(\xi_b(x; q)), \quad (6)$$

where $\phi_b : \mathbb{R}^3 \times \mathbb{R}^{n_q} \rightarrow \mathbb{R}$ returns the signed distance from x to the represented geometry of link b at configuration q .

The smooth Bernstein representation provides spatial distance gradients, and its composition with forward kinematics makes $\phi_b(x; q)$ differentiable with respect to q , enabling geometric clearance to be incorporated into optimization-based control and the geometric CBF construction discussed in Section III.

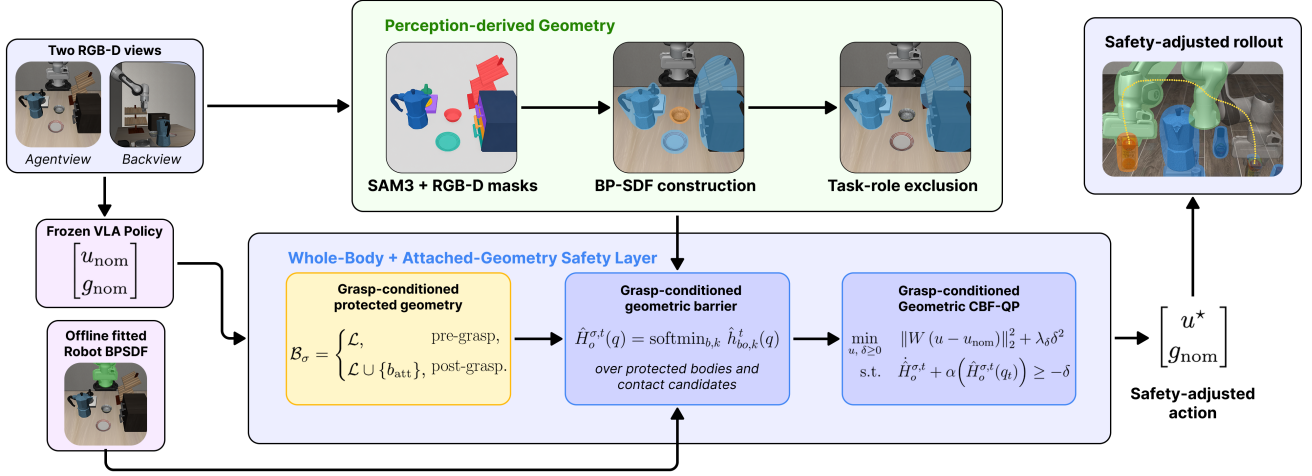


Fig. 2. Overview of WBAG. RGB-D observations reconstruct scene geometry, while robot BP-SDFs are fitted offline. Task-role exclusion defines scene obstacles, and the protected set expands from the articulated robot to include attached geometry after grasp confirmation. The resulting composite clearances are enforced by the grasp-conditioned geometric CBF-QP.

C. Problem Formulation

Let π denote a frozen VLA policy, instantiated in our experiments with $\pi_{0.5}$ [1], that maps an observation o and language instruction ℓ to a nominal action

$$a_{\text{nom}} = \pi(o, \ell) = \begin{bmatrix} u_{\text{nom}} \\ g_{\text{nom}} \end{bmatrix} \in \mathbb{R}^7, \quad (7)$$

where

$$u_{\text{nom}} = \begin{bmatrix} \Delta p \\ \Delta r \end{bmatrix} \in \mathbb{R}^6 \quad (8)$$

is the operational-space motion command. Here, $\Delta p \in \mathbb{R}^3$ and $\Delta r \in \mathbb{R}^3$ denote the translational and axis-angle rotational increments, respectively, and $g_{\text{nom}} \in \mathbb{R}$ is the scalar gripper command.

Let $q \in \mathbb{R}^{n_q}$ denote the robot joint configuration and $\mathcal{L}$ the set of modeled moving robot bodies. Let $\sigma \in \{0, 1\}$ denote the grasp mode, with $\sigma = 0$ before grasp confirmation and $\sigma = 1$ after grasp confirmation. Let b_{att} denote the attached target body, and define the grasp-conditioned protected set

$$\mathcal{B}_\sigma = \begin{cases} \mathcal{L}, & \sigma = 0, \\ \mathcal{L} \cup \{b_{\text{att}}\}, & \sigma = 1. \end{cases} \quad (9)$$

Thus, the set of protected bodies changes from the articulated robot to the robot-object system after attachment.

Let $\mathcal{O}$ denote the set of reconstructed scene bodies subject to collision-avoidance constraints. Objects requiring intentional task contact are excluded from $\mathcal{O}$, and the attached target is not constrained against itself. The protected bodies in $\mathcal{B}_\sigma$ and scene obstacles in $\mathcal{O}$ are represented using BP-SDFs.

Given $(u_{\text{nom}}, q, \sigma, \mathcal{O})$, our **objective** is to compute a step-wise safety-adjusted command $u^* \in \mathbb{R}^6$ that minimally deviates from u_{nom} while maintaining collision-free operation between every protected body $b \in \mathcal{B}_\sigma$ and every scene obstacle $o \in \mathcal{O}$. The executed action is

$$a^* = \begin{bmatrix} u^* \\ g_{\text{nom}} \end{bmatrix} \in \mathbb{R}^7. \quad (10)$$

The safety filter therefore modifies only the operational-space motion command while preserving the VLA's gripper command.

III. GRASP-CONDITIONED GEOMETRIC BARRIER

We first derive the barrier using ideal closest-point evaluation. Section IV then introduces the finite-contact approximation used online.

A. Pairwise BP-SDF Clearance

For protected body $b \in \mathcal{B}_\sigma$, let $\phi_b(x; q)$ denote its world-frame BP-SDF from (6). For obstacle $o \in \mathcal{O}$, let $\phi_o(x)$ denote its BP-SDF and define the offset obstacle level set

$$\Gamma_o = \{x \in \mathbb{R}^3 : \phi_o(x) = \ell_o\}, \quad (11)$$

where $\ell_o > 0$ is the obstacle offset.

For a given robot joint configuration q , define

$$x_{bo}^*(q) = \arg \min_{x \in \Gamma_o} \phi_b(x; q). \quad (12)$$

Here $x_{bo}^*(q)$ is the point on the obstacle offset surface Γ_o with minimum BP-SDF distance to protected body b . The corresponding pairwise clearance barrier is

$$h_{bo}(q) = \phi_b(x_{bo}^*(q); q) - \ell_b = \min_{x \in \Gamma_o} \phi_b(x; q) - \ell_b, \quad (13)$$

where $\ell_b \geq 0$ is the protected-body margin. Thus, $h_{bo}(q) \geq 0$ enforces the prescribed geometric separation between protected body b and obstacle o .

Following the closest-point gradient result in [11], assume that the relevant protected-body and obstacle BP-SDF level sets are convex, that $x_{bo}^*(q)$ varies smoothly with q , and that the corresponding BP-SDF gradients are nonzero on the level-set boundaries. The pairwise clearance gradient is then

$$\nabla_q h_{bo}(q) = \nabla_q \phi_b(x_{bo}^*(q); q). \quad (14)$$

B. Grasp-Conditioned Composite Barrier

The pairwise barrier h_{bo} measures clearance for one protected-body–obstacle pair. To enforce clearance for all protected bodies with a single constraint per obstacle, we aggregate the pairwise barriers using a soft minimum:

$$H_o^\sigma(q) = -\frac{1}{\beta} \log \left(\sum_{b \in \mathcal{B}_\sigma} e^{-\beta h_{bo}(q)} \right), \quad \beta > 0. \quad (15)$$

where β is the soft-min sharpness parameter, with larger values providing a tighter approximation to the minimum. Since $H_o^\sigma(q) \leq \min_{b \in \mathcal{B}_\sigma} h_{bo}(q)$, the condition $H_o^\sigma(q) \geq 0$ implies $h_{bo}(q) \geq 0$ for every $b \in \mathcal{B}_\sigma$.

We therefore define the grasp-conditioned safe set

$$\mathcal{C}_\sigma = \bigcap_{o \in \mathcal{O}} \{q : H_o^\sigma(q) \geq 0\}. \quad (16)$$

C. Action-to-Motion Mapping and Fixed-Mode Safety

For a fixed grasp mode σ , let $u \in \mathbb{R}^6$ denote an operational-space control command. We locally relate u to joint motion by

$$\dot{q} = G(q)u, \quad (17)$$

where $G(q) \in \mathbb{R}^{n_q \times 6}$ is the local action-to-motion mapping.

To derive the corresponding control constraint, differentiating the soft-min barrier in (15) gives

$$\begin{aligned} \nabla_q H_o^\sigma(q) &= \sum_{b \in \mathcal{B}_\sigma} w_{bo}(q) \nabla_q h_{bo}(q), \\ w_{bo}(q) &= \frac{e^{-\beta h_{bo}(q)}}{\sum_{b' \in \mathcal{B}_\sigma} e^{-\beta h_{b'o}(q)}}. \end{aligned} \quad (18)$$

Using (17), its time derivative is therefore

$$\dot{H}_o^\sigma(q) = \nabla_q H_o^\sigma(q)^\top G(q)u. \quad (19)$$

The corresponding CBF constraint thus becomes

$$\nabla_q H_o^\sigma(q)^\top G(q)u + \alpha(H_o^\sigma(q)) \geq 0, \quad \forall o \in \mathcal{O}. \quad (20)$$

Proposition 1 (Fixed-mode grasp-conditioned safety):

Fix $\sigma \in \{0, 1\}$ and suppose the convexity and regularity assumptions of (14) hold. If $q(0) \in \mathcal{C}_\sigma$ and (20) is satisfied for every $o \in \mathcal{O}$, then $\mathcal{C}_\sigma$ is forward invariant under (17).

Proof: For fixed σ , the protected set $\mathcal{B}_\sigma$ is fixed, and each h_{bo} is continuously differentiable under the convexity and regularity assumptions of (14). Hence, the soft-min composition H_o^σ is continuously differentiable. Equation (20) is therefore the standard CBF condition for the set $\{q : H_o^\sigma(q) \geq 0\}$. Enforcing it for every $o \in \mathcal{O}$ renders their intersection $\mathcal{C}_\sigma$ forward invariant. Since $H_o^\sigma \leq \min_{b \in \mathcal{B}_\sigma} h_{bo}$, membership in $\mathcal{C}_\sigma$ also implies nonnegative clearance for every represented protected-body–obstacle pair. ■

Remark 1 (Grasp-mode transition): Proposition 1 applies only within a fixed grasp mode. At attachment, the protected set expands from $\mathcal{B}_0$ to $\mathcal{B}_1$ by including the grasped object. The corresponding safe set therefore introduces additional clearance constraints involving the attached object. Although the robot configuration is continuous through the grasp transition, a configuration in $\mathcal{C}_0$ may not belong to $\mathcal{C}_1$ if

the newly attached object is already too close to another scene object. Thus, the proposition does not provide a safety guarantee across the discrete grasp transition itself. If the robot configuration at attachment lies in $\mathcal{C}_1$, the fixed mode forward invariance guarantee applies after attachment.

IV. WBAG SAFE CONTROL FRAMEWORK

Section III defines a grasp-conditioned geometric barrier that assumes exact continuous closest-point evaluation between each protected body and obstacle.

WBAG instantiates this formulation as the perception-to-control safety pipeline shown in Fig. 2. At episode initialization, multi-view RGB-D observations are used to reconstruct scene and target geometry, while robot-link BP-SDFs are precomputed offline. During execution, the protected set is updated when a grasp is confirmed and the attached-object geometry is propagated with the robot hand. The ideal closest-point barriers are approximated online using finite local contact search, and the resulting geometric constraints are mapped through a local action-to-motion model to the VLA's six-dimensional operational-space command. WBAG then solves a quadratic program (QP) that minimally modifies the nominal VLA action while satisfying the grasp-conditioned geometric CBF constraints.

A. Perception-derived Scene and Target Geometry

To instantiate the geometric barriers in Section III, WBAG combines precomputed robot geometry with perception-derived scene and target geometry. Robot-link BP-SDFs are precomputed offline from collision meshes using axis-wise second-difference constraints on the Bernstein coefficients to promote convex BP-SDF level sets. At episode initialization, scene and target geometry are reconstructed from multi-view RGB-D observations, as shown in Fig. 2. SAM3 [12] provides object masks for the camera observations, and the corresponding masked depth observations are fused into world-frame point clouds. Task-role information identifies the target, goal, and support objects and determines which reconstructed components are excluded from the obstacle set.

For each reconstructed object, including the task target, we construct a convex hull of the observed points and fit a BP-SDF using the same convexity-promoting constraints. The retained non-task objects define the scene obstacle set $\mathcal{O}$, while the target BP-SDF is retained for incorporation into the protected set after grasping.

B. Grasp-Conditioned Attached Geometry

To realize the grasp-conditioned protected set $\mathcal{B}_\sigma$ from Section III, WBAG detects attachment and estimates the target pose relative to the robot hand. Grasp confirmation is determined from gripper closure and target proximity over consecutive control steps. Upon confirmation, we set $\sigma = 1$; opening the gripper resets $\sigma = 0$. Let W , H , and A denote the world, hand, and attached-object model frames, respectively, and let $T_{XY} \in SE(3)$ denote the rigid

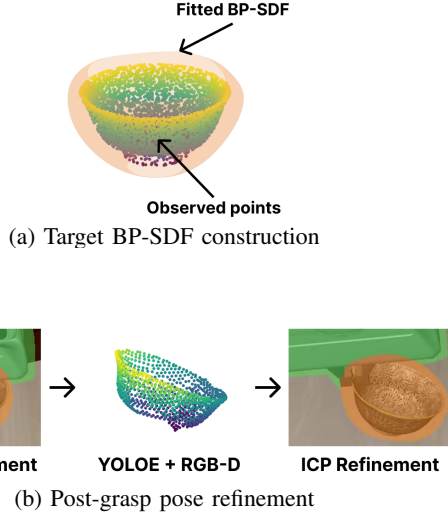


Fig. 3. Attached-geometry construction and refinement. (a) The task target is reconstructed from RGB-D observations and represented by a fitted BP-SDF. (b) After grasp confirmation, target RGB-D points are registered to the initial model with ICP, refining the hand-relative attachment pose before rigid propagation with the gripper.

transform from a frame Y to another frame X . At grasp time t_g , the hand-relative attachment transform is initialized as

$$T_{HA}^{t_g} = T_{WH}(q_{t_g})^{-1} \hat{T}_{WA}^{t_g}, \quad (21)$$

where $\hat{T}_{WA}^{t_g}$ is the perception-derived target pose obtained from the reconstruction in Section IV-A. The object's pose relative to the hand can change immediately after grasping as it settles within the gripper, since contact with the gripper and the object's geometry may induce a post-grasp reorientation. WBAG therefore refines the attachment transform during a short post-grasp window. YOLOE-26 [13] is used for repeated target segmentation during this window because it provides computationally efficient segmentation suitable for online inference, and the corresponding RGB-D points are registered to the initial target model using ICP [14] to update T_{HA} . After refinement, T_{HA} is held fixed and the attached geometry is propagated rigidly with the hand according to

$$T_{WA}(q_t) = T_{WH}(q_t) T_{HA}. \quad (22)$$

This propagated pose determines the world-frame BP-SDF of the attached object. Analogous to the robot-link representation in (6), we define

$$\phi_{\text{att}}(x; q) = \bar{\phi}_{\text{att}}(\xi_{\text{att}}(x; q)), \quad (23)$$

where $\xi_{\text{att}}(x; q)$ denotes the coordinates of the world-frame query point x expressed in the attached-object frame induced by $T_{WA}(q) = T_{WH}(q) T_{HA}$. Once grasped, the target becomes the protected body $b_{\text{att}} \in \mathcal{B}_1$ as defined in (9), and its BP-SDF enters the pairwise and composite barriers of Section III in the same manner as the robot-link BP-SDFs.

C. Online Geometric Barrier Approximation

Section III defines h_{bo} through a continuous closest-point search over Γ_o , which is impractical to solve for every protected-body-obstacle pair at each control step. WBAG therefore approximates this search using a finite set of local contact candidates. At control step t , let $\{x_{bo,k}^t\}_{k=1}^{N_{bo}^t}$ denote the candidate contact points for protected-body-obstacle pair (b, o) , where N_{bo}^t is the number of candidates and $x_{bo,k}^t \in \mathbb{R}^3$ is the k th candidate point. For each candidate, we define

$$\hat{h}_{bo,k}^t(q) = \phi_b(x_{bo,k}^t; q) - \ell_b, \quad (24)$$

where ℓ_b is the protected-body margin from (13). The candidate points are obtained by locally searching the obstacle level set Γ_o for regions of small protected-body clearance. The deployed composite barrier for obstacle o is then

$$\hat{H}_o^{\sigma,t}(q) = -\frac{1}{\beta} \log \left(\sum_{b \in \mathcal{B}_o} \sum_{k=1}^{N_{bo}^t} \exp \left[-\beta \hat{h}_{bo,k}^t(q) \right] \right), \quad (25)$$

using the same soft-min sharpness parameter β as in (15).

D. Action-to-Motion Mapping

The barrier in Section III is defined over robot configuration, whereas the frozen VLA outputs the six-dimensional operational-space increment u in (8). WBAG therefore maps this operational-space command to the local joint motion induced by the existing operational-space controller. Let $J(q) \in \mathbb{R}^{6 \times n_q}$ denote the geometric Jacobian and define its damped pseudoinverse as

$$J_\lambda^\#(q) = J(q)^\top (J(q)J(q)^\top + \lambda^2 I)^{-1}, \quad (26)$$

where $\lambda > 0$ is the damping coefficient. Let $D \in \mathbb{R}^{6 \times 6}$ contain the translational and rotational action scales of the downstream operational-space controller, and let $T_a > 0$ denote the effective action horizon. We instantiate the local action-to-motion map $G(q)$ in (17) as

$$\dot{q} = G_{\text{OSC}}(q)u, \quad G_{\text{OSC}}(q) = J_\lambda^\#(q) \frac{D}{T_a}. \quad (27)$$

Thus, $G_{\text{OSC}}(q) \in \mathbb{R}^{n_q \times 6}$ maps the VLA's operational-space increment to an approximate local joint velocity. Applying this map to the deployed barrier from Section IV-C gives

$$\hat{H}_o^{\sigma,t} = \nabla_q \hat{H}_o^{\sigma,t}(q_t)^\top G_{\text{OSC}}(q_t)u. \quad (28)$$

The modeled gripper geometry is treated as rigidly attached to the hand. For simplicity, finger motion induced by the scalar gripper command g_{nom} is not included in this local action-to-motion model.

E. Grasp-Conditioned Geometric CBF-QP

WBAG enforces the deployed grasp-conditioned CBF constraints through a minimum-intervention QP that selects the feasible command closest to the nominal VLA action:

$$\begin{aligned} (u^*, \delta^*) &= \arg \min_{u, \delta \geq 0} \|W(u - u_{\text{nom}})\|_2^2 + \lambda_\delta \delta^2 \\ \text{s.t.} \quad &\hat{H}_o^{\sigma,t} + \alpha \left(\hat{H}_o^{\sigma,t}(q_t) \right) \geq -\delta, \quad \forall o \in \mathcal{O}, \\ &u_{\min} \leq u \leq u_{\max}. \end{aligned} \quad (29)$$

TABLE I
EXPERIMENTAL PROGRESSION FROM EEf-MVEE REFERENCES TO SCENE-AWARE EEf PROTECTION AND WBAG.

Method	Perception	Obstacle geometry	Protected robot geometry	Attached geometry representation
AEGIS baseline [6]	Grounding DINO [15]	single MVEE	EEf ellipsoid	handcrafted EEf proxy
SAM3 EEf-MVEE	SAM3 [12]	single MVEE	EEf ellipsoid	handcrafted EEf proxy
SAM3 Scene EEf-MVEE	SAM3 [12]	scene MVEEs	EEf ellipsoid	handcrafted EEf proxy
WBAG EEf-only + attached geometry	SAM3 [12]	scene BP-SDFs	EEf-only	attached BP-SDF
WBAG w/o attached geometry	SAM3 [12]	scene BP-SDFs	whole-body BP-SDF	none
WBAG	SAM3 [12]	scene BP-SDFs	whole-body BP-SDF	attached BP-SDF

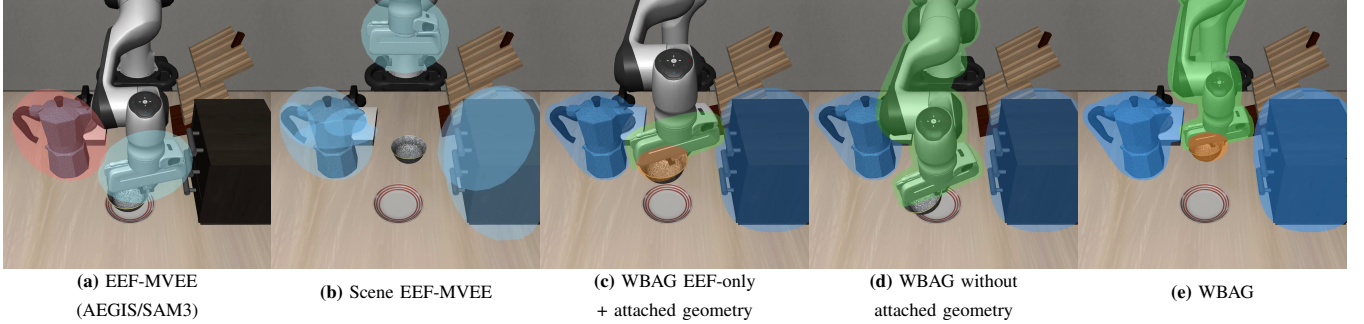


Fig. 4. Visual comparison of the configurations in Table I: (a) designated-obstacle EEf-MVEE, (b) scene-wide EEf-MVEE, (c) EEf-only with attached geometry, (d) whole-body without attached geometry, and (e) full WBAG.

Here, $W \geq 0$ weights deviation from the nominal VLA command, $\delta \geq 0$ is a shared slack variable, $\lambda_\delta > 0$ penalizes CBF relaxation, and $u_{\min}$ and $u_{\max}$ are the native action bounds. For $\delta = 0$, the constraint has the CBF form of (20), applied to the finite contact barrier $\hat{H}_o^{\sigma,t}$. Because deployment uses this approximation and permits $\delta > 0$, Proposition 1 applies to the exact closest-point formulation with zero-slack, whereas the deployed filter instead uses finite contact search and permits slack to maintain QP feasibility. The resulting u^* is executed with the unchanged gripper command as in (10).

V. RESULTS

A. Evaluation Setup

We evaluate on SafeLIBERO [6], a safety oriented extension of LIBERO [16] for tabletop manipulation. SafeLIBERO contains 16 tasks drawn from the LIBERO Goal, Long, Object, and Spatial suites, with four tasks from each suite. The Spatial tasks vary the spatial arrangement of the manipulated objects, the Object tasks vary which object must be transported, the Goal tasks vary the desired manipulation outcome, and the Long tasks require longer sequential interactions involving multiple objects. Across these suites, the robot performs pickup, transport, placement, articulated interactions such as opening a drawer, and multi object manipulation.

SafeLIBERO augments each task with two safety obstacle placements. In Level I, the designated obstacle is placed near the target object. In Level II, it is placed farther from the target but along the robot’s motion path. Each task and safety level contains 50 randomized episodes, yielding 1,600 episodes in total.

Following SafeLIBERO’s displacement based safety proxy [6], we evaluate all methods using a scene evaluator

that monitors every eligible scene body not required for task contact. An episode is unsafe if any such body is displaced by more than 0.01 m from its settled initial position $p_i(0)$:

$$\|p_i(t) - p_i(0)\|_1 > 0.01 \text{ m.} \quad (30)$$

Target objects, goals, and required support objects are excluded because contact with them may be necessary for task completion.

We report three episode level metrics. **Scene Safety** holds when no evaluated body violates Eq. (30). **Safe Success** denotes task completion with Scene Safety, whereas **Unsafe Success** denotes task completion without Scene Safety.

All configurations use the same frozen $\pi_{0.5}$ checkpoint [1] and identical episode initializations. Table I summarizes the evaluated configurations, and Fig. 4 visualizes their protected and obstacle geometries. *Policy Only* executes the frozen VLA without a safety filter. *AEGIS baseline* uses Grounding DINO [15] to localize the designated SafeLIBERO obstacle and represents safety using an end effector minimum volume enclosing ellipsoid (MVEE). *SAM3 EEf MVEE* replaces Grounding DINO with SAM3 [12] while retaining the same designated obstacle and end effector representation. *SAM3 Scene EEf MVEE* extends obstacle coverage to all eligible scene objects while retaining end effector protection. *WBAG EEf only + attached geometry* uses scene BP-SDFs and explicit attached object geometry but protects only the end effector. *WBAG without attached geometry* uses whole body robot BP-SDFs but omits the grasped object from the protected geometry. WBAG combines scene BP-SDFs, whole body robot protection, and explicit attached object geometry.

To isolate the safety representation from semantic identification errors, all methods receive simulator provided identities for the target, goal, and support objects when

TABLE II
PERFORMANCE ON SAFE LIBERO, AGGREGATED OVER LEVELS I AND II.

Suite	Metric	Policy only	AEGIS baseline	SAM3 EEf-MVEE	SAM3 Scene EEf-MVEE	WBAG EEf + attached	WBAG w/o attached	WBAG
Overall	Scene Safety	21.25%	70.87%	72.94%	82.75%	68.40%	77.50%	97.38%
	Safe Success	19.00%	51.06%	55.56%	46.25%	46.96%	46.31%	59.38%
	Unsafe Success	39.56%	11.25%	8.13%	4.31%	17.58%	14.69%	0.68%
Goal	Scene Safety	23.75%	83.50%	86.25%	77.25%	74.00%	88.75%	98.00%
	Safe Success	19.50%	72.75%	76.75%	52.25%	47.50%	53.25%	58.00%
	Unsafe Success	40.25%	7.00%	8.50%	8.25%	10.00%	6.75%	0.25%
Long	Scene Safety	22.25%	67.50%	64.75%	79.00%	62.00%	67.75%	96.25%
	Safe Success	20.00%	26.50%	30.50%	18.75%	34.00%	29.00%	48.75%
	Unsafe Success	36.25%	7.50%	7.75%	5.50%	21.00%	21.00%	2.25%
Object	Scene Safety	29.50%	59.50%	65.25%	90.00%	62.50%	71.00%	100.00%
	Safe Success	28.00%	47.50%	48.00%	46.25%	43.00%	45.00%	66.75%
	Unsafe Success	32.50%	21.50%	10.00%	1.25%	25.00%	23.25%	0.00%
Spatial	Scene Safety	17.75%	73.00%	75.50%	84.75%	69.25%	82.50%	95.25%
	Safe Success	17.00%	57.50%	67.00%	67.75%	59.25%	58.00%	64.00%
	Unsafe Success	44.50%	9.00%	6.25%	2.25%	12.75%	7.75%	0.25%

applicable. These identities replace the semantic identification stage that would otherwise determine which objects require task contact and should be excluded from the obstacle set. For *AEGIS baseline* and *SAM3 EEf MVEE*, simulator metadata provides the identity of the designated SafeLIBERO safety obstacle. This identity is used as the Grounding DINO [15] query for *AEGIS baseline* and as the SAM3 [12] prompt for *SAM3 EEf MVEE*. Thus, both methods localize the same designated obstacle from the visual observation and differ only in the grounding model. *SAM3 Scene EEf MVEE* and WBAG configurations instead use the fixed SAM3 prompt set {`bottle`, `can`, `box`, `plate`, `container`, `black notebook`, `cup`, `blue moka pot`, `black handle`} to localize eligible scene objects.

We instantiate the local action to motion map using controller scaling $D = \text{diag}(0.05, 0.05, 0.05, 0.5, 0.5, 0.5)$ and effective action horizon $T_a = 0.35$ s. We use obstacle margin $\ell_o = 0.01$ m and protected body margin $\ell_b = 0$. The soft minimum temperature, QP deviation weight, and slack penalty are $\beta = 100 \text{ m}^{-1}$, $W = \text{diag}(1, 1, 1, 10, 10, 10)$, and $\lambda_\delta = 10^4$, respectively. WBAG protects all modeled moving robot links, including the gripper and fingers, and adds the reconstructed target geometry after grasp confirmation. The EEf MVEE configurations use the handcrafted EEf proxy geometry from AEGIS [6], with fixed offset $[0, 0, -0.08]$ m and ellipsoid axes $[0.06, 0.12, 0.11]$ m by default, and axes $[0.06, 0.12, 0.20]$ m for orange juice, milk, or alphabet soup. Controller and evaluation settings are otherwise held fixed across configurations, and BP-SDF fitting and contact search settings are fixed across the WBAG configurations.

B. Benchmark Performance

Table II reports task and safety performance on SafeLIBERO. Scene Safety increases from 70.87% for *AEGIS baseline* to 72.94% for *SAM3 EEf MVEE*, 82.75% for *SAM3 Scene EEf MVEE*, and 97.38% for full WBAG. Extending obstacle coverage from *SAM3 EEf MVEE* to *SAM3 Scene*

EEf MVEE improves Scene Safety, but reduces Safe Success from 55.56% to 46.25%. WBAG further increases Scene Safety to 97.38% while achieving 59.38% Safe Success and reducing Unsafe Success to 0.68%. These results show that broader scene coverage alone improves safety at the cost of task completion, while WBAG improves safety without the same loss in Safe Success.

Across individual suites, WBAG achieves the highest Scene Safety and lowest Unsafe Success in all four suites. WBAG also achieves the highest Safe Success on Long and Object, while the EEf MVEE configurations achieve the highest Safe Success on Goal and Spatial. This variation indicates that the effect of the safety filter on task completion depends on the manipulation tasks within each suite.

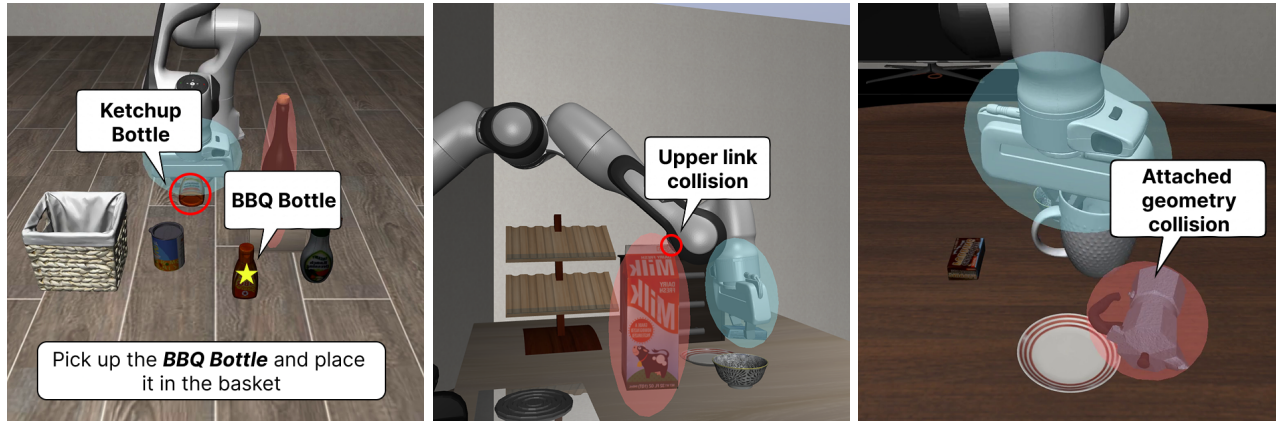
C. Ablation and Qualitative Analysis

The WBAG ablations in Table II isolate the effects of whole body and attached object protection. Comparing WBAG with *WBAG EEf only + attached geometry* shows that whole body protection improves both Scene Safety and Safe Success. Comparing WBAG with *WBAG without attached geometry* shows an additional benefit from explicitly protecting the grasped object. Together, these comparisons show that whole body and attached geometry modeling provide complementary benefits for collision avoidance.

Figure 5 illustrates the corresponding geometric failure modes. Incomplete scene coverage can leave non task objects unconstrained. End effector only protection can remain satisfied while an upstream robot link collides with an obstacle. A handcrafted end effector proxy can also fail to capture the shape and grasp relative pose of the manipulated object, allowing collisions involving attached geometry.

D. Runtime Analysis

WBAG requires 26.76 ms per filtering step on average. Contact search accounts for 20.41 ms, or approximately 76% of the complete filter runtime, while robot BP-SDF evaluation and QP optimization require only 0.26 ms and 1.58 ms, respectively. The primary computational bottleneck



(a) AEGIS baseline: incomplete scene coverage

(b) AEGIS baseline: upper-link collision

(c) AEGIS baseline: attached-geometry collision

Fig. 5. Representative failure modes of simplified safety geometry: (a) unintended pickup of a non-task object excluded from the designated-obstacle constraint set, (b) upper-link collision under EEF-only protection, and (c) attached-object collision under a handcrafted EEF proxy.

TABLE III

MEAN PER-STEP RUNTIME FOR WBAG AND POLICY INFERENCE.

Component	Mean (ms)
Contact search	20.41
Robot BP-SDF eval.	0.26
QP	1.58
Complete WBAG filter	26.76
Policy request	15.43

Per-step runtime is computed within each episode and then averaged across episodes. WBAG component timings are not exhaustive.

is therefore the geometric contact search used to approximate the closest point barriers rather than barrier evaluation or QP optimization.

VI. CONCLUSION

In this paper, we proposed WBAG, an inference-time safety framework for VLA manipulation that explicitly models whole-body and grasp-conditioned attached geometry. WBAG constructs BP-SDF clearance barriers over a grasp-conditioned protected set and enforces the resulting geometric CBF constraints directly on the VLA’s native operational-space action, without retraining the VLA or modifying the operational-space controller.

On SafeLIBERO, WBAG achieves 97.38% Scene Safety and 59.38% Safe Success. Controlled ablations show complementary benefits from whole-body and attached-geometry protection, while the complete filter requires 26.76 ms per filtering step on average. Together, these results support explicitly modeling the full robot–attached-object system rather than relying on end-effector-centered safety geometry for runtime VLA manipulation.

The fixed-mode forward-invariance result applies to the ideal geometric barrier, whereas the deployed framework uses finite contact search, a local action-to-motion approximation, and QP slack. Our evaluation also relies on simulator-provided task-role identities and does not explicitly model uncertainty in reconstructed or attached geometry. Future work will investigate uncertainty-aware geometry and task-role inference, dynamic obstacles, policy-aware filtering that better preserves task completion, with validation in the real world.

REFERENCES

- [1] K. Black *et al.*, “ $\pi_{0.5}$: A vision-language-action model with open-world generalization,” in *Proceedings of the 9th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 305. PMLR, 2025, pp. 17–40.
- [2] J. Fan, W. Xu, O. Sokolsky, I. Lee, and F. Kong, “SafeVLA-Bench: A benchmark for the success–safety gap in vision-language-action models,” *arXiv preprint arXiv:2606.00773*, 2026.
- [3] R. Cui *et al.*, “LIBERO-Safety: A comprehensive benchmark for physical and semantic safety in vision-language-action models,” *arXiv preprint arXiv:2606.23686*, 2026, accepted by ECCV 2026.
- [4] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [5] B. Zhang, Y. Zhang, J. Ji, Y. Lei, Y. Cai, J. Dai, Y. Chen, and Y. Yang, “SafeVLA: Towards safety alignment of vision-language-action model via constrained learning,” *arXiv preprint arXiv:2503.03480*, 2025.
- [6] S. Hu, Z. Liu, S. Liu, J. Cen, Z. Meng, S. Wang, X. Li, and X. He, “VLSA: Vision-language-action models with plug-and-play safety constraint layer,” in *2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026, arXiv:2512.11891.
- [7] S. Park, F. Zhang, B. Mirzasoleiman, S. Talebi, and N. Sehatbakhsh, “Your model already knows: Attention-guided safety filter for vision-language-action models,” *arXiv preprint arXiv:2606.09749*, 2026.
- [8] W. English, H. Zheng, and R. Ewetz, “Neuro-symbolic safety guidance for vision-language-action models via constrained flow matching,” *arXiv preprint arXiv:2607.01378*, 2026.
- [9] K. Sinaei, H.-C. Wu, and D. Ebeigbe, “Safe vision language action models via barrier enhanced flow matching,” *arXiv preprint arXiv:2607.29569*, 2026.
- [10] Y. Li, Y. Zhang, A. Razmjoo, and S. Calinon, “Representing robot geometry as distance fields: Applications to whole-body manipulation,” in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 15 351–15 357.
- [11] S. Jo, Y. Zhang, Y. Yang, and W. Luo, “Geometry-aware control barrier functions for collision avoidance via bernstein polynomial approximations,” in *2026 IEEE International Conference on Robotics and Automation (ICRA)*, 2026, arXiv:2605.30696.
- [12] N. Carion *et al.*, “SAM 3: Segment anything with concepts,” *arXiv preprint arXiv:2511.16719*, 2025.
- [13] R. Sapkota and M. Karkee, “YOLOE-26: Integrating YOLO26 with YOLOE for real-time open-vocabulary instance segmentation,” *arXiv preprint arXiv:2602.00168*, 2026.
- [14] P. J. Besl and N. D. McKay, “A method for registration of 3-d shapes,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 14, no. 2, pp. 239–256, 1992.
- [15] S. Liu *et al.*, “Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection,” in *Computer Vision – ECCV 2024*, 2024, pp. 38–55.
- [16] B. Liu, Y. Zhu, C. Gao, Y. Feng, Q. Liu, Y. Zhu, and P. Stone, “Libero: Benchmarking knowledge transfer for lifelong robot learning,” *arXiv preprint arXiv:2306.03310*, 2023.